

UNIVERSIDAD EAFIT
School of Applied Sciences and Engineering
Department of Mechanical Engineering
IM0240 – Machine Design I

Drawing package

Tripteron CNC router machine

Delivery 2 – English cover and drawing index

Group 2

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1. About this package

This package contains the complete set of manufacturing and assembly drawings of the Tripteron CNC router. The drawing sheets that follow this index are the CAD sheets as issued from SolidWorks; they are reproduced unchanged so that the dimensions, tolerances and revision state shown on them remain the ones the parts were made to.

General notes applying to every sheet:

- **All dimensions are in millimetres (mm).** Angles are in degrees.
- Projection: first-angle (ISO-A).
- Sheet format: A3 unless stated otherwise on the sheet.
- Unless otherwise indicated: linear tolerance ± 0.5 mm, angular tolerance $\pm 0.5^\circ$, break all sharp edges.
- Do not scale the drawing; work from the stated dimensions.
- Materials are stated in the title block of each sheet; the structural members are steel tube and plate, the links are $25 \times 25 \times 2$ mm square steel tube.

2. Drawing index

Sheet	Title	Drawing no.	Scale	Drawn by	Status
Group 1 – Chassis and structure					
1	Structural table – general drawing	ME-001	1:10	D. Zuluaga Henao	redrawn
2	Sheet metal plate – table top	ME-002	1:5	D. Zuluaga Henao	redrawn
3	Top support bracket	ME-003	1:2	D. Zuluaga Henao	redrawn
4	Motor support	ME-004	1:2	D. Zuluaga Henao	redrawn
5	Pulley plate – quarter sheet	ME-005	1:2	D. Zuluaga Henao	redrawn
6	Motor side support	ME-006	1:1	D. Zuluaga Henao	redrawn
7	Top support bracket 2	ME-007	1:1	D. Zuluaga Henao	redrawn
8	Support bracket 2	ME-008	1:2	D. Zuluaga Henao	redrawn
Group 2 – Electrical system					
9	Electrical box cover plate	ME-009	1:2	D. Zuluaga Henao	redrawn
10	Wiring diagram – motor control system	EE-001	N/A	D. Zuluaga Henao	redrawn
11	Motor control schematic	EE-001	N/A	D. Zuluaga Henao	redrawn
12	Electrical bill of materials	EE-002	N/A	D. Zuluaga Henao	redrawn
13	Stepper motor driver wiring diagram	EE-003	N/A	D. Zuluaga Henao	redrawn
Group 3 – Guide-supported axis system					
14	CNC router – guide assembly	ME-001	1:3	J. M. Rodríguez	as issued
15	Guide 1 – general views	ME-002	1:4	J. M. Rodríguez	as issued
16	Guide 1 – exploded view and component list	ME-003	1:3	J. M. Rodríguez	as issued
17	Guide rail assembly	ME-003	1:2	G. Riaño	as issued
18	Guide 2 – general views	ME-004	1:3	G. Riaño	as issued
19	Guide 3 – general views	ME-005	1:3	G. Riaño	as issued
20	Guide carriage – isometric view	–	2:1	G. Riaño	as issued

Sheet	Title	Drawing no.	Scale	Drawn by	Status
21	Guide carriage – views	ME-006	1:2	G. Riaño	as issued
22	Bearing support – views	ME-007	1:1	J. M. Rodríguez	as issued
Group 4 – Arms (R-pair system)					
23	Arm assembly – component list	–	1:2	R. Bueno	as issued
24	Elbow and bearing assembly – section A–A	–	2:1	D. Zuluaga Henao	as issued
25	Bearing housing tube Ø30, gauge 16	–	2:1	R. Bueno	as issued
26	Tube Ø30, gauge 16	–	2:1	R. Bueno	as issued
27	Arm – fish-mouth cut, 1 in, 90°	–	1:2	R. Bueno	as issued
28	Milling tool support	–	2:1	D. Amell Osorio	as issued
29	Elbow assembly 1 – section A–A and component list	–	1:10	D. Zuluaga Henao	as issued
Group 5 – Assembled machine					
30	Complete machine – general views	–	1:10	G. Riaño Barrera	as issued
31	Complete machine – exploded view and part list	–	1:10	D. Amell Osorio	as issued
32	Washer (spacer)	–	10:1	D. Zuluaga Henao	as issued
33	Internal bearing spacer	–	5:1	D. Zuluaga Henao	as issued
34	Bearing housing tube	–	2:1	D. Zuluaga Henao	as issued
35	Solid tube with bore	–	2:1	D. Zuluaga Henao	as issued

Status says how each sheet reaches this package. *Redrawn* means the sheet was drawn again with an English title block and English notes, keeping the geometry and the dimensions of the original; *as issued* means the sheet is reproduced exactly as it was submitted on 3 March 2024, with its Spanish title block. Thirteen of the thirty-five sheets were redrawn; the rest are untouched.

The “drawn by” column reproduces the title block of each CAD sheet. Sebastián Rincón Martínez carried the static analysis of Delivery 3 together with Juan Manuel Rodríguez Duque, and also contributed to the concept selection of Delivery 1.

3. Relation to the other deliveries

Group	Where it is verified
Chassis and structure	Delivery 3, Sheet 5: finite element analysis of the welded frame (safety factor 66, maximum deflection 0.102 mm).
Guide-supported axes	Delivery 3, Sheet 3: lead-screw torque, efficiency, stresses and column buckling (safety factor 3.21 against buckling), and Sheet 4: bushing loads and rod deflection of the carriages (safety factor 1.87, the tightest in the machine).
Arms	Delivery 3, Sheet 2: internal loads and bending stress in the six links (maximum 26.1 MPa, safety factor 9.6).
Assembled machine	Delivery 3, Sheet 1: kinematic model, stroke usage and workspace (a 200 mm cube, 8.6 dm ³).
Electrical system	Delivery 1: motor selection (NEMA 34 86HB250, 4.5 N m); Delivery 3, Sheet 3: required torque is 7.1 % of that.

Note on the sheets that follow. The sheets marked *as issued* above keep their original Spanish title blocks. The reading key is: *Materia* = course, *Proyecto* = project, *Dibujó* = drawn by, *Verif.* = checked, *Aprob.* = approved, *Escala* = scale, *Unidades* = units, *Cantidad* = quantity, *Formato* = sheet size, *Material* = material, *Hoja 1 de 1* = sheet 1 of 1, *Las cotas se expresan en mm* = all dimensions in millimetres, *No cambie la escala* = do not scale the drawing, *Romper aristas* = break sharp edges.